

C O D E W A V E 2 0 2 4

M I S S I O N C T R L



Communicating
Distress Signal is
Vital



What if the signal
needs to be
discreet



We use
Gesture

Missing Girl Is Rescued After Using Hand Signal From TikTok

The girl flashed the hand signal from a car on a Kentucky interstate, the authorities said. It was created as a way for people to indicate that they are at risk of abuse and need help.



HAND SIGNAL
TO ASK FOR HELP

Family > Family News

**A distress hand signal sequence
shared on TikTok could save your
child's life**

GESTURES



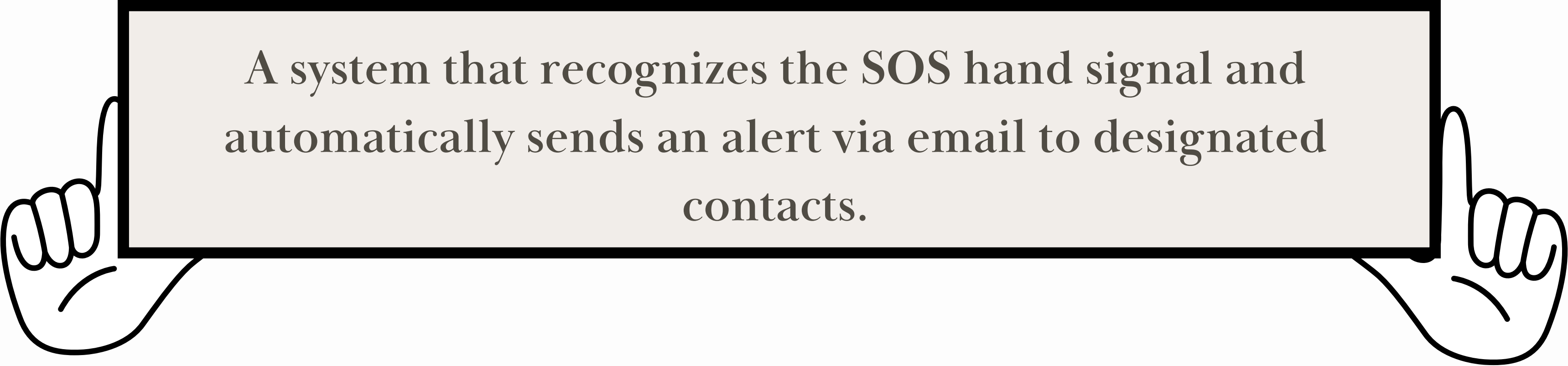
Is the gesture seen?

Does the signal reach the right
authority



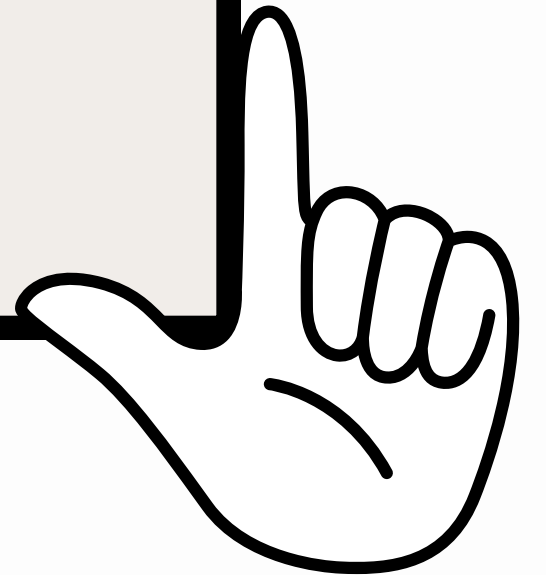
SIGNALBUDDY

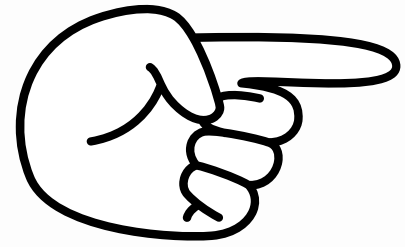
MISSION CTRL



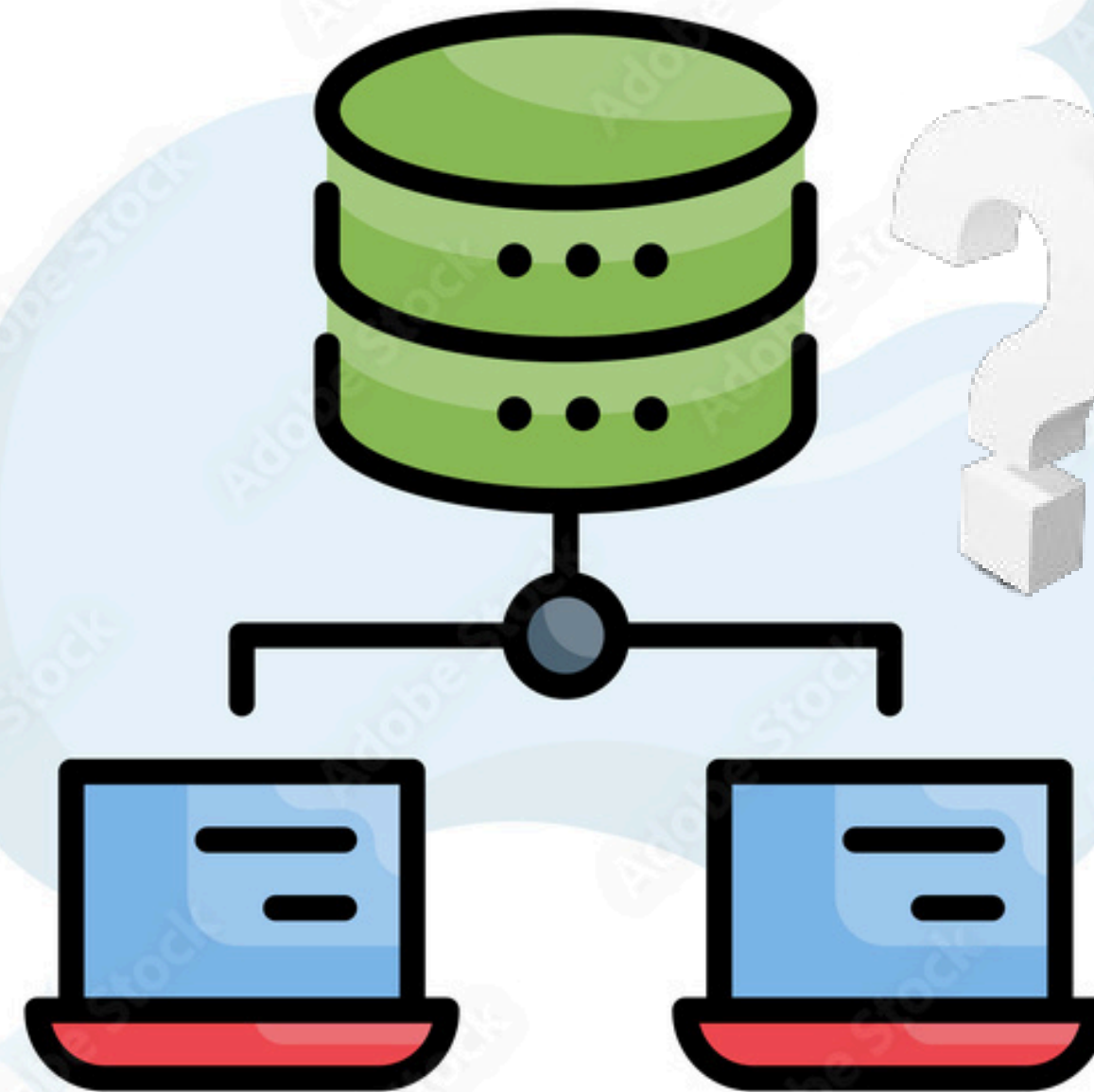
A system that recognizes the SOS hand signal and automatically sends an alert via email to designated contacts.

By utilizing AI, it enhances safety and ensures timely assistance when traditional communication methods are unavailable.



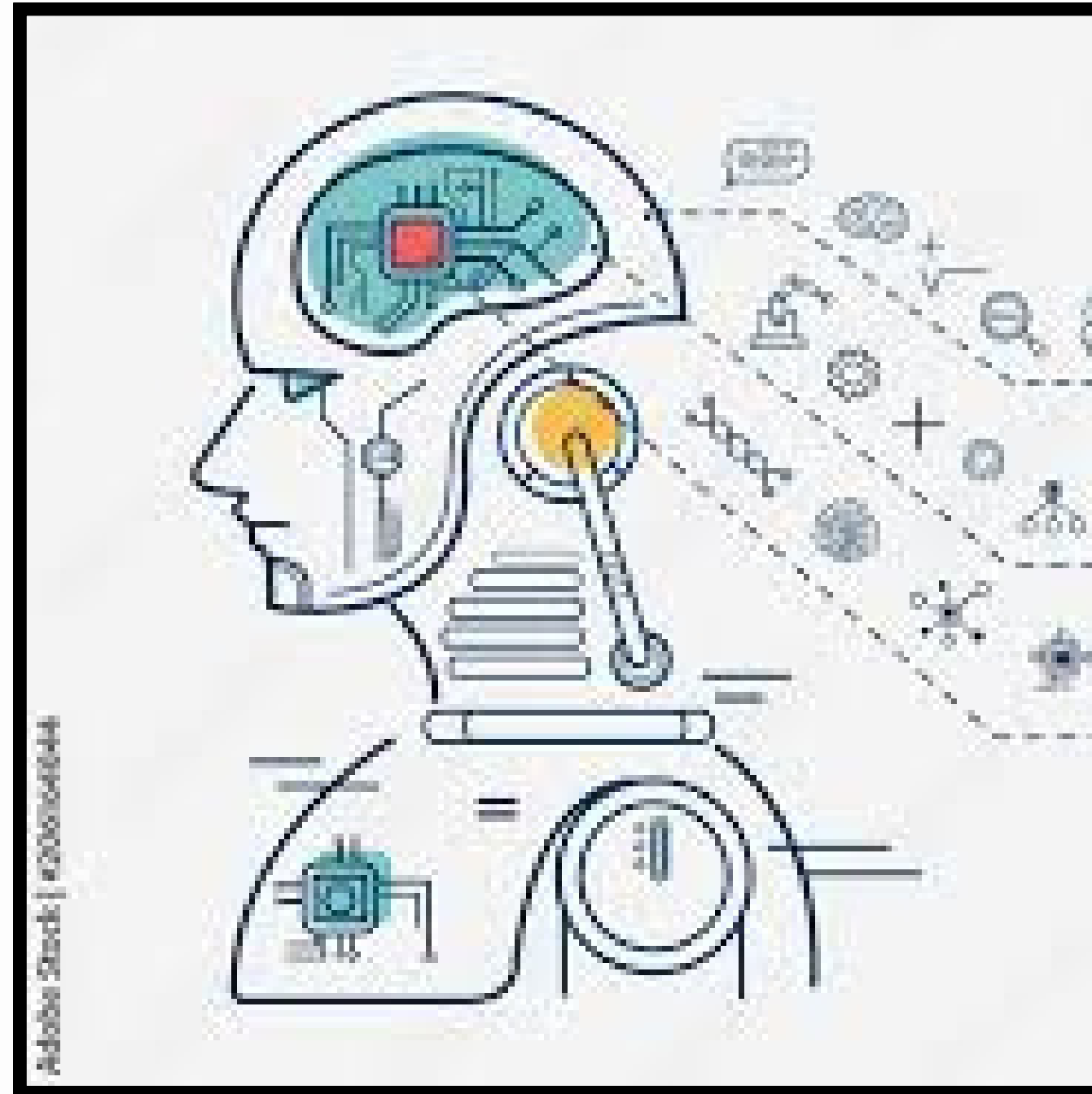


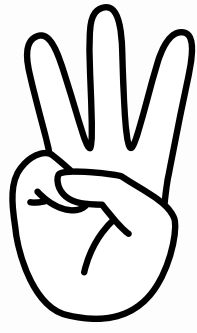
DATASET



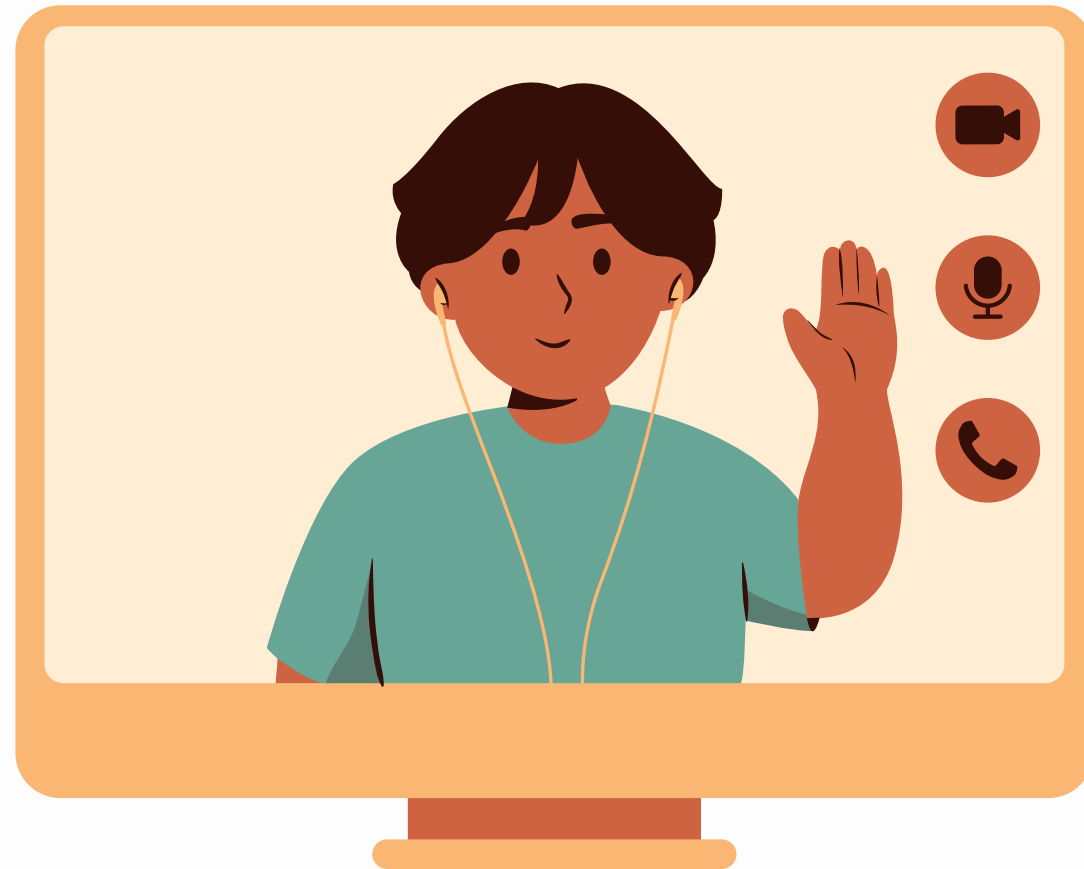


DEVELOP A MODEL





DEPLOY THE MODEL





CREATE A TRIGGER RESPONSE



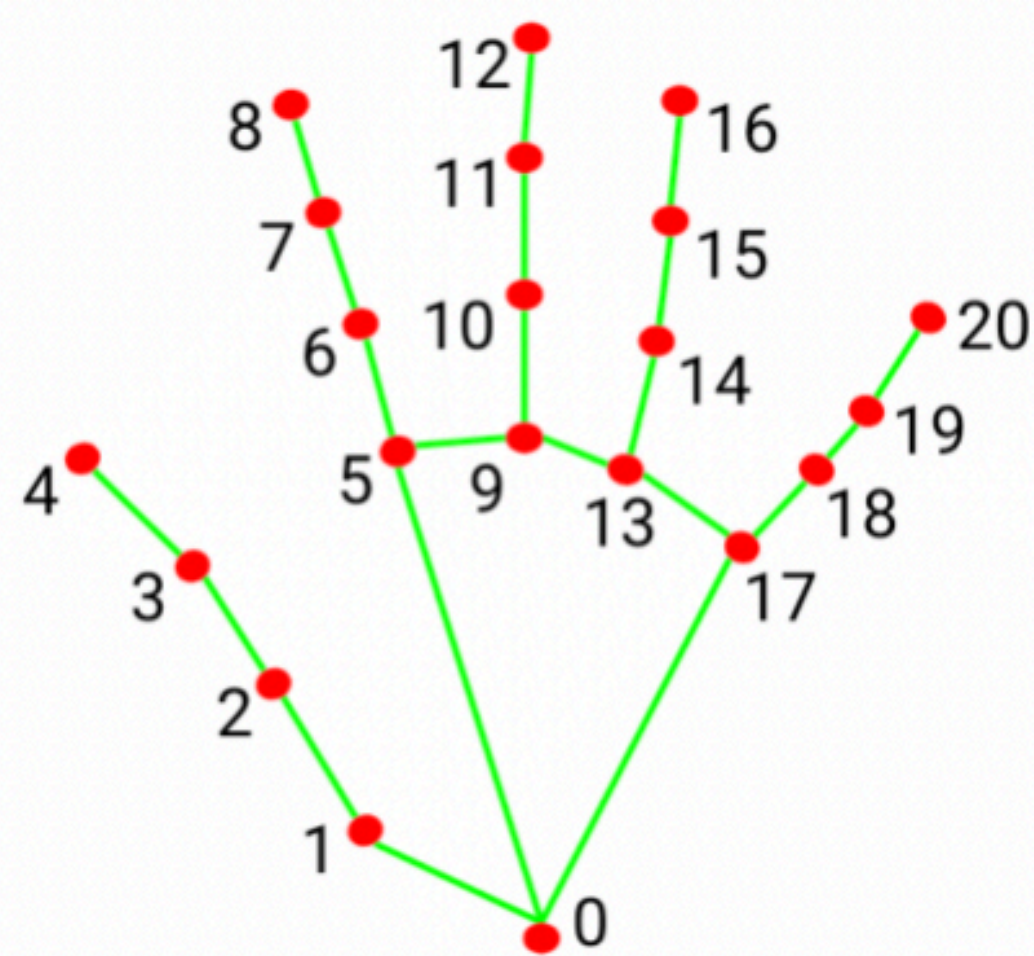
Application





Hand landmark model

By Google MediaPipe Solutions

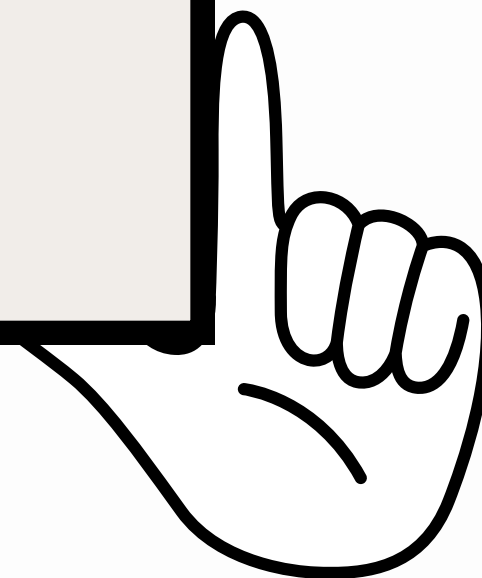
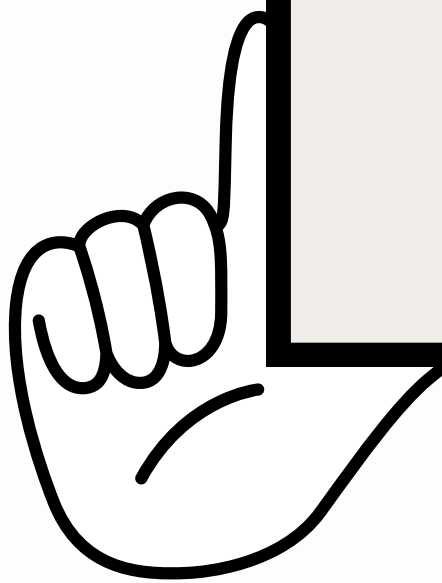


- 0. WRIST
- 1. THUMB_CMC
- 2. THUMB_MCP
- 3. THUMB_IP
- 4. THUMB_TIP
- 5. INDEX_FINGER_MCP
- 6. INDEX_FINGER_PIP
- 7. INDEX_FINGER_DIP
- 8. INDEX_FINGER_TIP
- 9. MIDDLE_FINGER_MCP
- 10. MIDDLE_FINGER_PIP

- 11. MIDDLE_FINGER_DIP
- 12. MIDDLE_FINGER_TIP
- 13. RING_FINGER_MCP
- 14. RING_FINGER_PIP
- 15. RING_FINGER_DIP
- 16. RING_FINGER_TIP
- 17. PINKY_MCP
- 18. PINKY_PIP
- 19. PINKY_DIP
- 20. PINKY_TIP



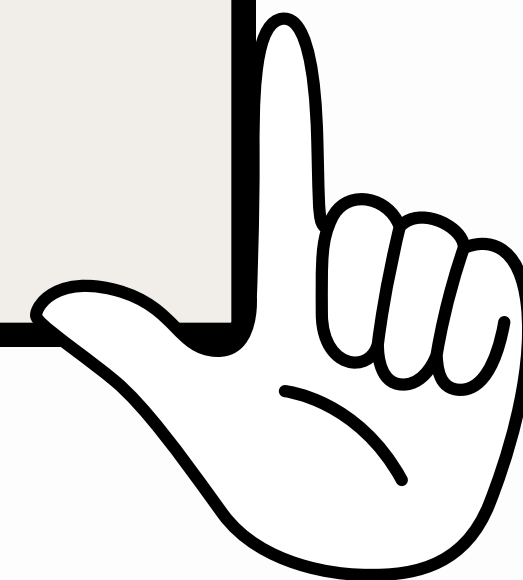
Activity Recognition





LSTM

(Long Short Term Memory)



Deployment



Future Development

For Future development of the project, we can

- Package the model as a stand-alone extension for web browser and real-time video conferencing software.
- Integrate the software with dedicated GPS hardware or maps for more accurate location services.

Thank You!

Sources

- [Victor, D., & Medina, E. \(2021, November 8\). Missing Girl Is Rescued After Using Hand Signal From TikTok. The New York Times.](#)
- [NATION, T. \(2022, May 11\). One signal that can save your life. Nationthailand; THE NATION.](#)
- [Maycock, S. \(2021, November 12\). A distress hand signal sequence shared on TikTok could save your child's life. GoodTo; GoodtoKnow.](#)
- [Hand landmarks detection guide. \(2024\). Google AI for Developers.](#)

Any Queries?

